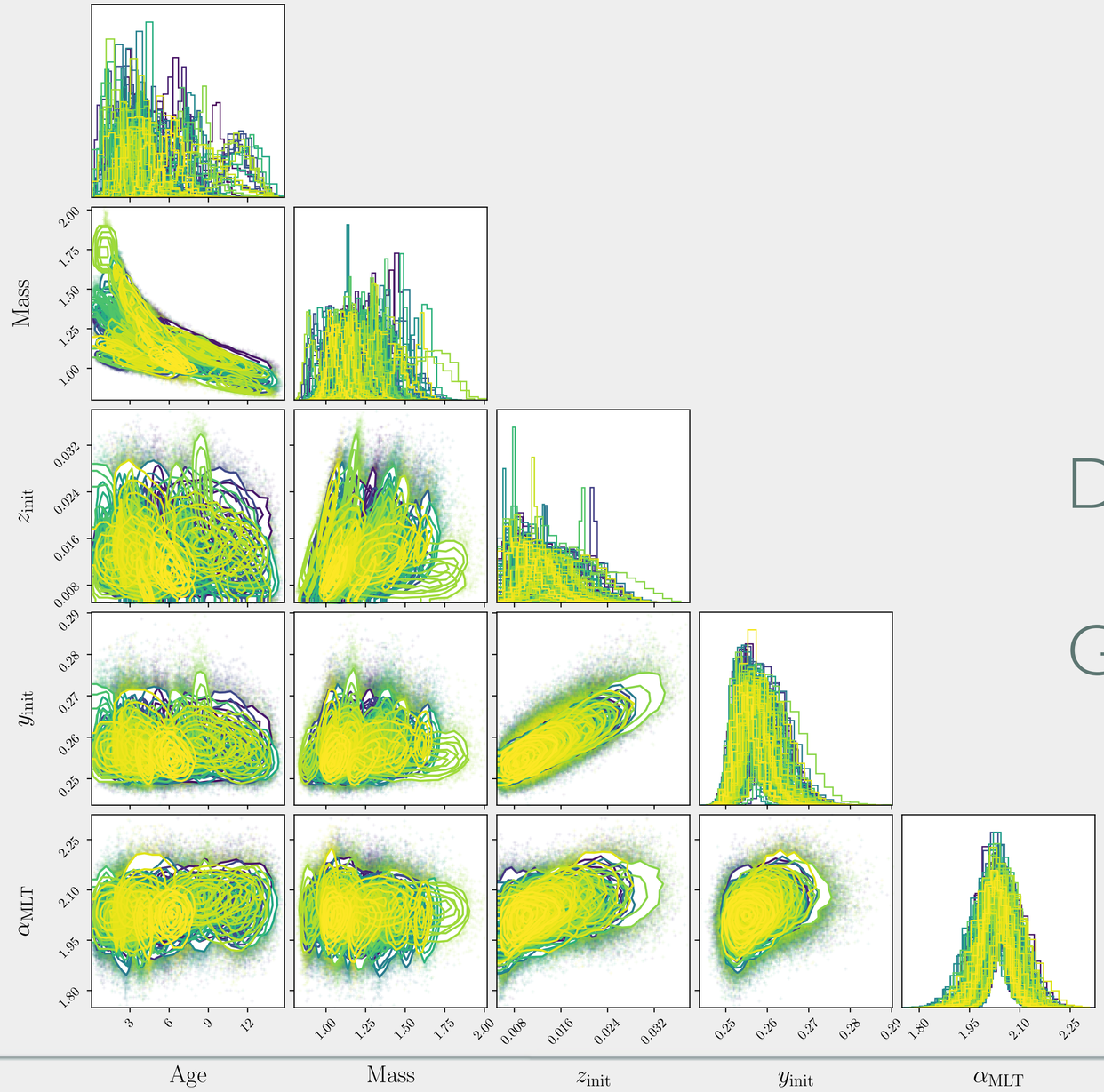
Inference from Scientific Data



Distributions Lecture

Guy Davies

Inference

- **Introduction** - Intro, Logic, Probability recap
- **Probability** - Distributions, Moments, Central Limit Theorem
- **Bayesian methods** - Parameter estimation, Model selection, Stochastic sampling
- **Frequentist methods** - Model fitting, Hypothesis testing
- **Fun stuff** - We'll see, but maybe Gaussian Processes, …



Inference -Week 2

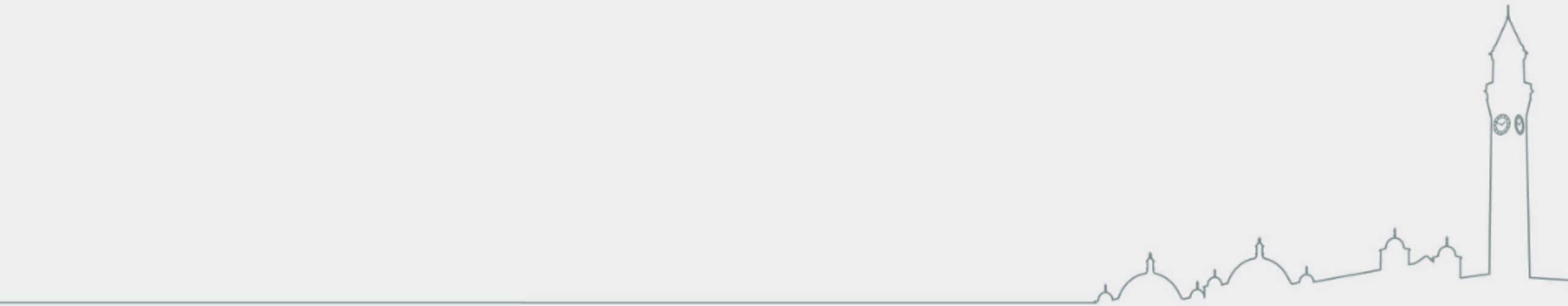
Today:

- Discrete distributions
- Continuous distributions
- The variance
- Moments
- ~~Moment Generating Functions~~
- ~~MGF Examples~~



Inference - Week 2

The distribution and the PDF are different things!



Inference - Discrete

Discrete distributions

We define all possible outcomes of an experiment:

$x_1, x_2, x_3, \ldots x_i, \ldots$

There could be an inifinte number of possible outcomes but the number of outcomes is countable.

This is then called the sample space.

The probability distribution has a probability mass function (PMF) and assigns the probability to each event,

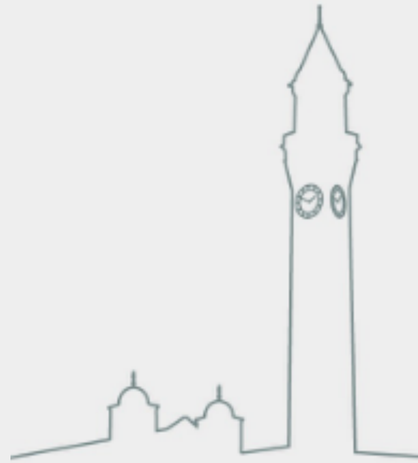
$P(x_i)$

And from the law of total probability,

$\sum_i P(x_i) = 1.$

By our definition from last week we can say,

$0 \leq P(x_i) \leq 1.$



Inference - Discrete

And usually it is convient to assign numbers to all elements of the sample space.

we can then define the mean (or expectation if you want),

$$\langle x \rangle = \sum_i x_i P(x_i).$$

Also, the expectation of any function of x_i, $f(x_i)$ is

$$\langle f \rangle = \sum_i f(x_i) P(x_i).$$



Inference - Continuous

Continuous distributions

Here the sample space x is a real number, for example,

$x \in (-\infty, \infty)$.

But now there are an uncountable number of possibilities! For a continuous distirbution we use the probability density function (PDF),

$$\text{Prob}(a < x < b) = \int_a^b \mathrm{d}x P(x),$$

which is then integrating over a probability density. $P(x)$ does not calculate a probability but does give us a way of to calculate probabilities.



Inference - The variance

The variance

We define the variance:

$$\mathrm{Var}(x) = \left\langle (x - \langle x \rangle)^2 \right\rangle ,$$

where $\langle\rangle$ is defining the expectation. We have defined the expectation before and can write out,

$$\mathrm{Var}(x) = \int \mathrm{d}x \, P(x) \, (x - \langle x \rangle)^2,$$

which could again be expanded but the notation will get tedious.

We define the standard deviation σ as

$$\sigma_x = \sqrt{\mathrm{Var(x)}}.$$

It is possible to derive and certainly useful to know that,

$$\mathrm{Var}(x) = \left\langle x^2 \right\rangle - \langle x \rangle^2.$$



Inference - Moments

Non-central moments

We define the non-central moment μ_k as:

$$\mu_k = \left\langle x^k \right\rangle,$$

and we will say this is the kth moment. This is very general definition but generally we will find use for the moments where $k = 0, 1, 2, 3, \ldots$

Let's look at some of the moments:

$$\mu_0 = \left\langle x^0 \right\rangle = \int \mathrm{d}x \, P(x) \, x^0 = 1,$$

which you probably assumed already. But, let's just take a moment (sorry) to recognise that this is a number. All the moments are numbers and are ways of describing a distribution.



Inference - Moments

Remembering that we use mean and expectation interchangeably,

$$\mu_1 = \langle x \rangle \,,$$

which is the mean!

$$\mu_2 = \left\langle x^2 \right\rangle = \int \mathrm{d}x \, P(x) \, x^2 \,,$$

which is related to the variance because

$$\mathrm{Var}(x) = \left\langle x^2 \right\rangle - \langle x \rangle^2 = \mu_2 - \mu_1^2 .$$

Remember how moments are just numbers? Well, these 'numbers' can be used to tell us all we need to know about a distribution. We do not need a PMF/PDF to describe a distribution.



Inference - Central Moments

Central moments

Central moments are a thing too,

$$\nu_k = \left\langle (x - \langle x \rangle)^k \right\rangle,$$

but probably focus on the non-central moments for the purposes of this course.



Inference - Moments

The relation between the non-central and central moments

Consider the relation between the central and non-central moments. Can we calculate one from the other?

$$\nu_k = \left\langle (x - \langle x \rangle)^k \right\rangle .$$

Do you remember the binomial expansion?

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k}\, y^k.$$

And can you use it?

$$\nu_k = \left\langle x^k - k \langle x \rangle x^{k-1} + \ldots + \langle x \rangle^k \right\rangle .$$

$$\nu_k = \left\langle x^k \right\rangle - k \langle x \rangle \left\langle x^{k-1} \right\rangle + \ldots + \langle x \rangle^k .$$

$$\nu_k = \mu_k - k \langle x \rangle \mu_{k-1} + \ldots + \langle x \rangle^k .$$

$$\nu_k = \mu_k - k \mu_1 \mu_{k-1} + \ldots + \mu_1^k .$$

This gives you a relation between the central and the non-central moments and perhaps a useful piece of information to get some marks.



Inference - Next time …

Distribution examples

The Central Limit Theorem

Quiet a lot of maths (sorry)

